

Perception of Symmetry of Actual and Modulated Self-Avatar Gait Movements During Treadmill Walking

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ABSTRACT

In virtual reality, it is possible to simulate one's visual self-representation by mapping one's body movements to those of an avatar. Accepting the virtual body as part of one's own body creates an ownership illusion. This study aimed to assess the perception threshold between a subject's actual gait movements and those of their modulated self-avatar during treadmill walking. Preliminary results on two subjects suggest that healthy subjects can detect the mismatch, but differences may exist between subjects.

Keywords: Self-avatar, virtual reality, ownership illusion, rehabilitation, visual feedback, gait symmetry.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities; H.5.2 [Information Interfaces and Presentation]: User Interfaces—Evaluation/Methodology.

1 INTRODUCTION

Over the years, virtual reality (VR) has emerged as a therapeutic tool for the assessment and treatment of patients with neurological conditions [1]. In immersive VR, self-avatars can replicate one's body movements in the virtual environment (VE) through virtual bodies (self-avatars) and substitute the real body, which is not visible inside the VE. If the visuomotor synchronization between the real and virtual movements is preserved, the user can experience a sense of ownership over the avatar. Previous evidence showed that altering the one-to-one mapping between subject and avatar can lead to motor changes in subjects during an arm movement task. The conflict between visual and proprioceptive signals led subjects to unconsciously alter their own arm movements towards those of the avatar to diminish the incongruency between the real and virtual bodies [2,3]. This adaptation phenomenon, also referred to as the "follower-effect" becomes particularly interesting for gait rehabilitation of post-stroke individuals who suffer from gait asymmetry. In such a context, the avatar's step length could diminish the appearance of a deficit or inversely, amplify it by exaggerating the gait asymmetry. The avatar's modified movements could implicitly affect the patients' movement to alter their step lengths and drive a more symmetrical pattern. In contrast to motor tasks in the upper limb (reaching and flexing), the follower-effect and the possible adaptation effects has been scarcely studied during gait. However, preliminary results on nine healthy subjects from our lab demonstrated a tendency to induce changes in stride length as a result of the modified self-avatar's stride length [5]. Nevertheless, to date, the physical limits of these unconscious motor adaptations remains unknown. Understanding the perception threshold between the actual movements of the user and that of his/her avatar can orient the future design of an avatar's step length modulations in real-time when aiming to induce corrections in the user. Therefore, this study aims to evaluate the perception threshold between subject's actual gait movements and that of their self-avatar when its step length is lengthened or shortened.

2 METHODOLOGY

2.1 Materials

Thirty healthy subjects will be recruited to volunteer for this study. So far two (2) subjects (one male, one female) have been recruited (31.5 ± 2.1 years; 1.82 ± 1.0 m; 82.4 ± 23.2 kg). Ethical approval was obtained through the CRCHUM ethics committee and both subjects signed written consent before the start of the experiment. Data collection are performed on an AMTI split-belt treadmill that measures ground reaction forces and moments. The subjects are fitted with a safety harness and 16 rigid body marker clusters tracked by a 12-camera optoelectronic system running Vicon Tracker 3.6, which measures kinematic data. Vicon Pegasus Advanced 1.2.2 (Vicon, Oxford, inc) is used for retargeting the body positions and orientations onto the virtual avatar. The avatar is personalized by taking the subjects' body height and body weight into account when generating it with MakeHuman 1.2.0 (MakeHuman). The VE is animated in Unity 3D game engine and replicates our research lab with a large virtual mirror placed in front of the subject. The VE is visualized through a head-mounted display (HMD), the HTC VIVE Pro Eye, with a first-person perspective (1PP) relative to the avatar (Figure 1).

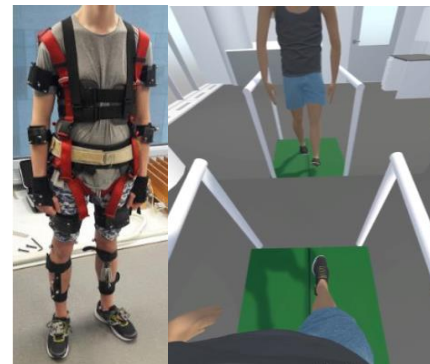


Figure 1: Left: A participant wearing mocap marker clusters and a security harness. Right: A subject's avatar walking in VE, shown in 1PP in front of the virtual mirror.

2.2 Modulations

An algorithm was developed which allows to create an asymmetry by gradually changing the step length of one of the avatar's legs in real-time. To be able to lengthen or shorten the step length of the avatar in real-time, the algorithm detects the minimum and maximum distance, between the ankle and hip joint of the avatar, in the anteroposterior axis. To preserve the visuomotor synchronization, the moment of the heel strike of the avatar always corresponds with that of the user, whatever the current modulation of step length. The Final IK (inverse kinematics solution) asset was used to solve the joints angular displacements of the avatar.

2.3 Protocol

The experiment started with a five minute habituation period without wearing the HMD to establish comfortable gait speed. Subjects were then asked to walk another five minutes with their self-avatar mimicking their gait kinematics. During this period, the baseline step length of each subject was measured. The perception threshold between the subject's actual gait movements and that of their self-avatars was assessed for both legs over four different conditions randomized across subjects. During the Dominant Lengthening (DL) and Non-Dominant Lengthening (NDL) conditions, the avatar's step length was increased by 1% of the baseline step length, after every ten steps of the modulated leg until the point of detection was verbally reported by the subjects. During the Dominant Shortening (DS) and the Non-Dominant Shortening (NDS) conditions the step length of the avatar was reduced by 1% of the baseline step length every ten steps until the detection was reported. Subjects were not aware which leg was modulated during the experimental trials. Each of the four conditions were evaluated using an ascending/descending series approach. Both the ascending and descending series were repeated 3 times. In the ascending series, the avatar's step length was gradually increased until the subjects reported that they perceived that the avatar was no longer accurately replicating their movements. In contrast, during the descending series the avatar was walking asymmetrically (the average asymmetry that was detected in the ascending series) from the start and the asymmetry was gradually decreased until the subjects reported that avatar was replicating their movements.

3 RESULTS

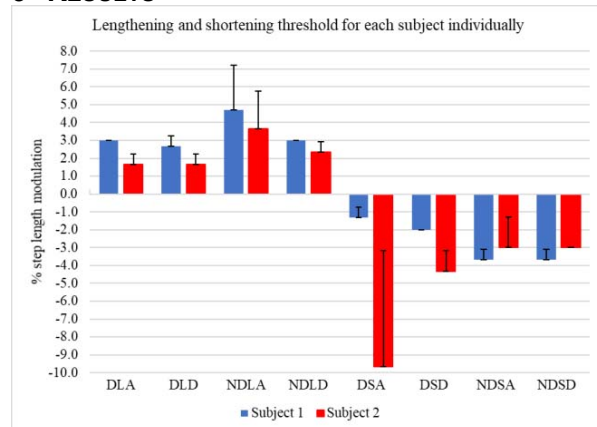


Figure 2: Mean and standard-deviation of the point of detection for each subject individually. DominantLengthening Ascending (DLA), DominantLengthening Descending (DLD), NonDominantLengthening Ascending (NDLA), NonDominantLengthening Descending (NDLD), DominantShortening Ascending (DSA), DominantShortening Descending (DSD), NonDominantShortening Ascending (NDSA), NonDominantShortening Descending (NDS).

Here we present the results for 2 subjects (gait speed: 1.1 m/s, 1.2m/s respectively). For each subject, the mean of the three points of detection for the ascending and descending thresholds was calculated for all four conditions.

Figure 2 illustrates the point of detection for each subject individually for the lengthening and shortening threshold, respectively. Both subjects were able to detect mismatch/match between themselves and their self-avatar in all four conditions. For the lengthening threshold, the detection point is different for both subjects but, similar patterns across conditions were observed. The non-dominant leg has a higher threshold (subject 1; $4.7\% \pm 2.5$, subject 2; $3.7\% \pm 3.7$), with asymmetry thus detected later then for the dominant leg (subject1; $3.0\% \pm 0.0$, subject 2; $1.7\% \pm 0.58$) in both subjects. The shortening threshold results shows a variability between subjects for the dominant leg (subject 1; $-1.3\% \pm 0.58$, subject 2; $-9.7\% \pm 6.5$) but similar results for the non-dominant leg (subject 1; $-3.7\% \pm 0.58$, subject 2; $-3.0\% \pm 1.7$) during the ascending series.

4 DISCUSSION

The purpose of this study was to evaluate the perception threshold between the subject's gait movements and their self-avatar when the step length of the avatar is lengthened or shortened. The results presented on 2 subjects showed that the point of detection in the lengthening threshold were similar. However, the shortening threshold shows more variety between subjects. Identifying the perception threshold of gait symmetry between a subject's movements and that of their modified avatar could provide a better understanding of the extent to which embodied healthy subjects can perceive their avatars as their own bodies. Further analysis will be conducted to evaluate if subjects adapted their step lengths and modified their gait symmetry before the point of detection during both thresholds. The perception threshold could be the minimum needed to elicit the "follower-effect", in other words, the minimum spatial offset between subject and avatar to be able to observe a motor adjustment while minimizing the sensory conflict.

5 CONCLUSION

The preliminary results of this study demonstrate that healthy subjects are able to detect the absence or presence of a mismatch in gait symmetry between themselves and their self-avatar, yet differences exist between subjects. The study is ongoing and the assessment of more subjects will allow to validate (or invalidate) these observations and document the normal threshold detection of perception step length between the avatar and subject's movements.

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